

ADVANCED CERTIFICATE OF EDUCATION S6 TERM ONE 2026

PHYSICS

Paper 1

3 hours

INSTRUCTIONS TO CANDIDATES

Answer **FOUR** questions.

Section A is compulsory, and attempt one item in each of the sections B and C.

Use provided data and support materials to guide your solutions.

Clearly show illustrations, calculations and assumptions.

SECTION A. (compulsory)

1. PART I (ITEM ONE): PARTICLES.

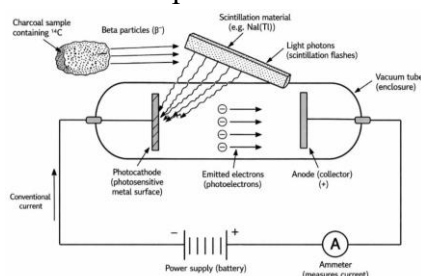
A team of History researcher is conducting an investigation to determine the period when a certain island was last occupied by human activity. Historic information is that the island was last occupied over 10,000 years ago. During a visit to the island, charcoal remains were discovered. An archeologist on the team got excited and assured his colleagues that the discovered charcoal remains can be analyzed to answer their question.

He told them that they can use carbon-14 dating since **carbon-14 is radioactive** by measuring activity of the sample. As they inquired to understand more, he told them that this process is **spontaneous and random**. They got more interested in understanding this process and how it can help to determine the period the island was last occupied.

To measure the activity of the sample, the laboratory uses scintillation-photoelectric detector system that works by photoelectric effect. The team does not know how such a photoelectric detector can measure activity of carbon-14. The detector can give close to accurate activity if the kinetic energy of electrons emitted from the cathode is greater than $6.0 \times 10^{-19} \text{ J}$. From the laboratory it was determined that the activity was 0.075 Bq. The researchers would like to know how accuracy of this result is affected and whether photoelectric effect has any other application in dairy life.

Support material

Scintillation-photoelectric detector used



Half-life of carbon-14 = 5730 years

Activity of carbon-14 in living sample = 0.3 Bq

Work function of photocathode = $3.2 \times 10^{-19} \text{ J}$

Frequency of emitted light = $1.5 \times 10^{15} \text{ Hz}$

Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$

TASK

As a physics student, you have been approached. Analyze the information and provide answers to the group of history researchers.

2. PART II (FORCES AND MOTION)

During many sports competitions, officiation errors usually cause arguments and most times cause fights. In recent times, there has been introduction of video technology to help in minimising human officiation errors.

In the final moments of a basketball game, a player begins to sprint from one end of the court when 4.0 s remain to the end of the game clock. The player runs in a straight line, starting from rest and accelerating uniformly towards the centre line, which is 14m horizontally from the hoop and immediately projects the ball on reaching that point. The ball enters the hoop after the buzzer (end of game signal/ whistle). The referee was not sure if the basket/ score should count, and decided to consult video replay.

A replay video captured the instant the player started sprinting from end of the court and the instant the ball entered the hoop but the instant when the ball left the players hands was not visible. However, the replay shows that the ball entered the hoop 5.2 s after the sprint began.

The ball was released from a height of 2.4m and passed through a hoop at a height of 305cm.

At the closing ceremony, David a former professional basketball player advised the players to concentrate at perfection of jump shots (where a player jumps up and holds a stationary position before releasing the ball), perfect smooth follow through (where a player throws the ball by extending arms pointing towards the target/hoop) and snap shots (where the fingers should create back spin on the basketball as its being released).

Support material:

- Acceleration due to gravity $g = 9.8ms^{-2}$
- The basketball was released at an angle of 46° to the horizontal with a velocity of $12ms^{-1}$.
- The basket counts only if the ball is released before the buzzer.
- Air resistance is negligible.

TASK: As a physics student, use the principles of mechanics to:

- (a) Determine whether the score is valid, analyse and explain the motion of both the player and the ball.
- (b) Explain the closing remarks of David.

SECTION B. (ENERGY) Attempt one item.

3. At a renewable energy research station near Lake Victoria, intern students are assisting Engineers to investigate whether lake waves can support a small-scale Wave power project.

The engineers explain that the available wave power device can only generate electricity if the wave amplitude is at least 0.25m. the students therefore need to determine whether the lake waves are suitable for the project.

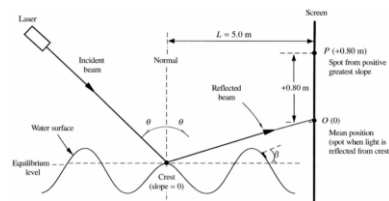
The lake wave is modelled by the equation, $y = A \sin(2\pi ft - kx)$, where f is the frequency of the wave and k is the wave number, $y(m)$ is displacement of the water waves from undisturbed surface, and $x(m)$ is the distance moved by the water waves. For future research purposes, the students would like to have a record of the wave equation of the water waves.

Because the direct measurement of the moving waves is difficult, the students decided to use a screen method based on reflection of light to determine amplitude. A narrow laser beam is directed on to the water surface; the reflected beam falls on a vertical screen placed 5.0m away. When light is reflected from the crest of the wave (inclination of reflecting surface is zero), it reaches mean position O on the screen. When light is reflected from the wave surface at its greatest positive slope (inclination of the reflecting surface is β), the reflected spot shifts 0.8m above the mean position O. The students need to use knowledge of light reflection to determine the inclination of greatest slope β of the wave.

To check the accuracy of the students' results, Engineers use a second method. A converging lens is placed in the path of the reflected beam so that the light is focused into an optical Fiber. The optical Fiber carries the light to a photodetector and the electronically recorded signal is compared with the students' screen measurements. The students do not know how the optical Fiber transmits the light and why a convex lens was used in this option by the Engineers.

Support material.

- Experimental setup for measuring angle of inclination of greatest slope of the wave.



- The angle of inclination of the greatest slope of the wave is equal to the maximum slope of the wave $(\frac{dy}{dx})_{max} = Ak = \beta$.
- Wavelength $\lambda = 20.0 \text{ m}$, and period $T = 5.0 \text{ s}$.
- Assume small angle conditions in radians apply.

Task: As a physics student:

- Analyse the information and help the students to find out if the project would be supported by the lake waves, and determine the lake wave equation.
 - Discuss the accuracy check method used by the Engineers to address students' concerns.
4. At a certain secondary school, the learners are developing a solar water heating system to provide warm water for laboratory use. They use a large concave mirror of surface diameter 2.4m to concentrate sunlight onto a blackened metal container filled with water. The light intensity incident on the earth's surface is $1.4 \times 10^3 \text{ W m}^{-2}$, only 65% of power reflected by the mirror reaches the blackened metal container. During the first trial, the water became only slightly warm. The learners suspected that the container is not placed at the correct position where the sunlight is most concentrated. The students did not have an idea on how to do this. After positioning the container, the students decided to put ice at -5° frozen from 5liters of water into the container. The water needed for laboratory use must be at a temperature not less than 80° . For the system to be considered efficient for continuous use, the ice should be transformed to water at 80° in a time not more than a single lesson period.

James, one of the students thereafter inquired from the teacher if there are other real-life applications of concave mirrors. The teacher informed him that the same mirrors are used by dentists for teeth examination. The teacher challenged him to research on how this application can be illustrated using a ray diagram and use the ray diagram to establish the relation between object and image distances from the mirror and the focal length of the mirror.

Support material:

- **Specific** heat capacity of ice = $2.1 \times 10^3 J Kg^{-1} K^{-1}$
- specific heat capacity of water = $4.2 \times 10^3 J Kg^{-1} K^{-1}$
- specific latent heat of fusion = $3.3 \times 10^5 J Kg^{-1}$
- a single lesson period is 40 minutes

Task:

- Help the students with the guidelines to properly position the container for optimal light intensity reception and analyse if the system would be efficient for continuous use.
- As a physics student approached by James, help him to deal with the research assignment.

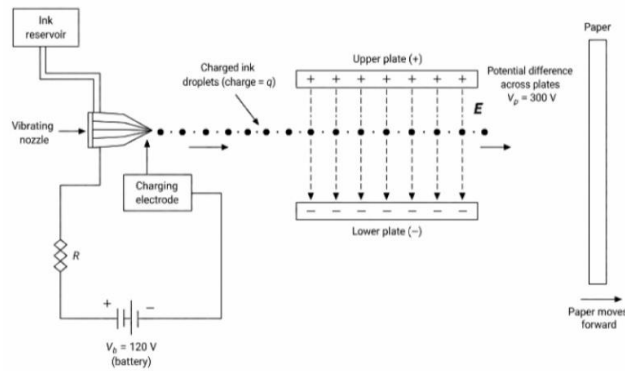
SECTION C. (CHARGES AND FIELD)

- Standard collage Ntungamo uses an inkjet printer to produce examination papers. The printer forms images by ejecting tiny ink droplets that are electrically charged. These droplets pass between two parallel plates at a potential difference of 300V where an electric field deflects them into the correct positions on the paper. For the ink droplets to be properly directed to the paper, the quantity of charge on the oppositely charged plates must be greater than $1.4 \times 10^{-7} C$ and force on each ink droplet should be more than $1.2 \times 10^{-6} N$.

The charging system is powered by DC supply of 120V through a resistor R to control current. According to specifications a of 0.15A flow through R for proper working of the charging system. However, during use, the printer becomes hot and produces faint and poorly aligned prints. A technician has been invited to resolve the issues, on opening the printer, he discovered that the resistor was spoilt and needed replacement but the exact resistor value was not reflected anywhere. The technician discovered that the resistor was made of Nichrome wire and decided to get the wire and improvise to make the required resistor but did not know the length of the wire to buy. It was not clear to the technician if there was a problem with the deflecting system of the printer.

Support material:

- Area of the deflecting plates = $5.0 \times 10^{-3} m^2$
- Separation of the plates = $1.0 \times 10^{-4} m$
- Charge on one droplet = $4.0 \times 10^{-13} C$
- Permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12} Fm^{-1}$
- Resistivity of Nichrome wire = $1.0 \times 10^{-6} \Omega m$
- Diameter of the nichrome wire to be used = $0.1 mm$
- Diagram of the printing system



TASK:

- (a) Explain how such a printer casts prints of the paper and analyse if the ink droplets are cast on the proper positions and in proper quantity portions and advise the technician.
 - (b) Help the technician to determine the length of wire needed to improvise to make the required wire.
6. A certain school has been experiencing a challenge of being supplied fake school laboratory equipment. The school procurement committee has come with a regulation of testing any equipment supplied to the school before it is accepted. A small Van de Graaff generator was supplied to the school; the generator uses a motor driven insulating belt to transfer charge to a spherical metal dome of radius 0.12m. The school laboratory technician did not know the system works and even what to check for. According to the operating manual, a maximum charge can be obtained on the dome if the motor driving the generator operates at 12V and draws a current of 0.5A, at least 0.05% of the electrical energy should be stored as electrostatic energy and that the spherical dome acts as an isolated spherical capacitor.

Davis a senior six physics student was requested to work with the laboratory Technician. Davis suggested that they can test for charge distribution and how much of the motor electrical energy is converted to stored electrostatic energy.

During the experiment, the total charge accumulated on the dome was measured to be $2.4 \times 10^{-6} \text{C}$ when it had been kept running for one minute. As the generator remained switched on, sparks were observed around the dome. This surprised both Davis and the technician. Before the two could complete the test, Davis was called for his next lesson.

Support material:

- Permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12} \text{Fm}^{-1}$
- Dielectric strength of air $= 3.0 \times 10^6 \text{Vm}^{-1}$.

TASK:

- (a) As a physics student, analyse the information, and help the laboratory technician to understand the working of the system and make a decision on whether to accept the generator or not.
- (b) Explain what could have caused the sparks around the dome, with clear details to convince Davis.

END